

SustainaFund: every decision and why

A \$100 million equity mandate built as a mixed-integer quadratic programme in FICO Xpress. This is the complete record of what the data was, what we did to it, which model we chose, why each parameter and constraint has the value it has, and what every test showed — including the tests that came back negative.

Universe 1,093 stocks → 1,077 optimised Prices 2015-01-01 → 2025-12-31, daily
Solver Xpress 9.9.1 / optimizer 47.01.02 Budget \$100,000,000

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RECOMMENDED BOOK	14.30% return	RETURN / RISK	1.346	HOLDINGS
30 of 1,077	10.62% risk			
	WEIGHTED ESG	70.00 exactly		

01 – THE MANDATE

What was actually asked for

A fictional fund, SustainaFund, wants to invest \$100 million in equities over a mid-term horizon. The task is to choose *which* stocks and *how much* of the budget goes into each, trading expected return against risk, while satisfying a set of concentration and sustainability conditions the brief fixes in advance.

Four conditions came with the brief and are not ours to choose. Quoting the case study description directly, the selected portfolio must satisfy:

CONDITION AS WRITTEN IN THE BRIEF	IN THE MODEL
Investment in any particular stock must not exceed 20% of the budget and must not be lower than 1%	$0.01 \leq w_i \leq 0.20$ if held
Total investment in any single region must not exceed 60% (so at least 40% goes to the other region)	$\sum w \text{ over region} \leq 0.60$
The number of different stocks purchased must be at least 30	$\sum y_i \geq 30$
The ESG score amongst selected stocks, weighted by fraction, must be at least 70	$\sum \text{ESG}_i \cdot w_i \geq 70$
There are exactly two regions in this universe — United States and Europe — which is why a 60% cap on one implies a 40% floor on the other. That arithmetic turns out to matter later, in section 09.	

Everything beyond those four is a decision we made and therefore have to justify. The distinction is kept sharp throughout this document, because claiming credit for a condition the brief handed us would be the easiest thing in the project to get caught on.

Three deliverables were required: the optimisation model itself, a dashboard with visualisation elements aimed at "the upper management team", and a presentation. The brief also asks that any external data be "documented and, ideally, reproducible" — a requirement that drove a good deal of the engineering in section 13.

02 — THE DATA WE GOT

Four files, and what was wrong with each

The dataset is built from real market data: **STOXX Europe 600 and S&P 500 constituents, sourced from the LSEG platform** — ten years of daily prices and annual ESG scores. What arrived:

FILE	SHAPE	CONTENTS AND DEFECTS
<code>stockprices_full.csv</code>	1,093 × 2,870	Daily closing prices, in local currency , 2015-01-01 to 2025-12-31. Scattered gaps; two series with undeclared corporate-action breaks.
<code>shares_full.csv</code>	1,093 rows	Ticker, region, country, ESG score. 49 ESG scores missing .
<code>sectors.xlsx</code>	1,093 rows	Sector per stock, no gaps.
<code>prices_lseg_dividend_adjusted.csv</code>	1,093 × 2,870	Arrived later: total-return prices (dividends reinvested). Six series entirely empty .

A second LSEG export arrived near the end of the project carrying refreshed ESG scores plus GICS sector and industry classifications. It was parsed, measured and deliberately *not* adopted; the reasoning is in section 11.

Why the universe shrinks from 1,093 to 1,077

Two separate filters, and it matters that they are reported separately rather than as one number, because a dashboard that shows either as the other is misleading:

- **Six stocks lost their price series** in the total-return export (`URW.PA`, `AVB.N`, `EQR.N`, `HOLN.S`, `EA.OQ`, `HWM.N`) — 2,870 of 2,870 values absent where the earlier file had a full history. Three are REITs, whose dividend-adjustment factors are the largest, so the extraction most likely failed on them. They were *dropped rather than back-filled* from the unadjusted file: an unadjusted series carries roughly 3 pp less return, and the optimiser would then penalise those stocks for a data defect rather than for their actual economics. 1,093 → 1,087.
- **Ten stocks fall below the per-stock ESG floor** of 30 (section 09). 1,087 → 1,077.

03 — CLEANING

Five cleaning decisions, each with a reason

1. Forward-fill gaps *after* a stock's first price, never before

A missing value after a stock has started trading is a non-trading day or a data gap, and carrying the last price forward is the standard, conservative treatment — it produces a zero return for that day rather than inventing a move. A missing value *before* a stock's first price is not a gap at all: the stock did not exist yet. Filling backwards would invent a price history and let the optimiser buy a company before it listed. 118 stocks in this universe list after January 2015, so this is not a marginal case.

2. Rescale London prices from pence to pounds

LSEG quotes London-listed stocks in **GBp (pence)**, not pounds. Left uncorrected, every UK stock's price level is inflated a hundredfold. Price *levels* cancel out of returns, so this would not corrupt μ or Σ — but it would corrupt the currency conversion in section 04, which divides by an FX rate quoted in pounds.

3. Truncate at trading discontinuities, detected on *local* prices

Two series contain breaks that are corporate actions, not market moves:

MEASURED — TWO DATA DEFECTS FOUND

ZEG.L · reverse takeover | BMPS.MI · recapitalisation

Detected by a rule rather than by eye: a run of ten or more identical consecutive prices — a frozen quote, typical of a suspended listing — immediately followed by a jump of more than 30%. Both were **truncated** (the history before the break discarded, the stock kept) rather than dropped entirely, so the company remains investable on its post-event history. Detection runs on local prices deliberately: doing it after currency conversion would let a genuine FX move trigger a false positive.

4. Impute the 49 missing ESG scores with the sector median

The alternative was dropping 49 stocks from a 1,093-stock universe over one missing field. A sector median is the least assumptive filler available — it says "behaves like its peers" rather than inventing a level. When the second LSEG export later supplied real values for 47 of the 49, we could finally check the imputation, and the honest answer is mixed:

MEASURED — THE IMPUTATION WAS INDIVIDUALLY POOR AND COLLECTIVELY HARMLESS

mean absolute error 14.07 ESG points · worst case 45.92 (BFT.WA: 62.03 imputed vs 31.46 actual)

Per stock the sector median was substantially wrong. The portfolio never noticed, because the optimiser was not leaning on those 49 names — using the real values moves the recommended book by **0.069 pp** of return. Robust, but by luck rather than by design, which is the strongest argument for eventually adopting the real values.

5. Drop the six empty series rather than substitute

Covered in section 02. The principle is worth stating on its own: **never fill a gap with a quantity measured on a different basis**. Mixing an unadjusted price into a total-return panel makes that stock look permanently worse by its dividend yield, and the optimiser reads a data artefact as an economic fact.

04 — WHY US DOLLARS

The conversion that changes the answer

Prices in this universe are quoted in eight currencies across nineteen exchanges. We convert everything to USD at daily European Central Bank reference rates. This is the single most consequential cleaning decision in the project, and it is also the one most often waved away — so here is the full argument.

The objection, and why it is wrong

The natural objection is that currency conversion cannot matter, because returns are *ratios*: dividing every price in a series by the same constant leaves the return series unchanged. That is true for a constant. An exchange rate is not a constant — it moves daily. Taking logs makes the consequence explicit:

```
# a price converted to USD is the local price divided by the FX rate  $\ln(P_{\text{usd}}) = \ln(P_{\text{local}}) + \ln(fx)$  # so the daily USD return carries an additive FX term that does NOT cancel  $r_{\text{usd}}(t) = r_{\text{local}}(t) + r_{\text{fx}}(t)$ 
```

The FX return survives into the return series. And because it is *shared* by every stock quoted in that currency, it is not noise — it is a common factor, exactly the kind of thing a covariance matrix exists to capture.

What it does to the two estimates, measured separately

The effect on expected returns and the effect on risk are completely different in size, which is why they have to be reported separately:

MEASURED — EFFECT ON EXPECTED RETURN

under 0.3 pp on μ

Small, and for a comprehensible reason: over ten years the currencies largely offset one another, so the average FX drift is close to zero. If you only looked at μ you would conclude the conversion was cosmetic.

MEASURED — EFFECT ON RISK, WHICH IS WHERE IT BITES

min-variance book: 8.46% risk believed vs 10.13% actually realised — a 19.7% understatement

A local-currency covariance matrix has no FX factor in it, so it cannot see that a Swiss and a German holding share a euro-franc exposure. The optimiser, trying to minimise variance, walks straight into that blind spot: it builds a book it believes carries 8.46% volatility and which in fact carries 10.13%. Understating risk is the one failure a risk-averse mandate cannot tolerate.

MEASURED — FX IS A GENUINE RISK FACTOR, NOT AN ARTEFACT

identical holdings realised 8.04% volatility in local currency vs 8.99% in USD
• $\text{var}(fx) = 17.6\%$ of total variance

Model-free: the same weight vector, valued two ways. Nearly a fifth of total portfolio variance is currency. A model that omits it is not "simplified", it is wrong about a sixth of the risk.

The economic argument, which is the real one

Beyond the statistics: a local-currency return is what a *domestic* investor earns. SustainaFund is a single fund with one reporting currency. If it buys a Swiss stock that rises 10% in francs while the franc falls 10% against the dollar, the fund earned approximately nothing. Optimising local-currency returns optimises a quantity no single investor actually receives.

HOW THE "+5.71 PP AT MATCHED RISK" FIGURE IS CONSTRUCTED

Worth stating precisely, because it is easy to describe loosely. In four steps:

1. Take the weight vectors from the frontier solved on *local-currency* estimates.
2. **Re-price each one under the USD estimates:** its true return is $\mu_{\text{usd}} \cdot w$ and its true risk is $\sqrt{(w^T \Sigma_{\text{usd}} w)}$. This is the step that produces "8.46% believed vs 10.13% actual" — same portfolio, measured correctly.
3. Solve a *new* frontier on the USD inputs.
4. For each old portfolio, take its **true** risk as the x-coordinate and read off the new frontier's return at that same risk by linear interpolation. The gain is the new frontier's return minus the old portfolio's *true* return.

So both sides of the comparison are measured in the same space — under the correct USD estimates — and differ only in which covariance the optimiser was given when it chose the weights. It is not comparing a believed number with an actual one. **+5.71 pp** is the gain at the matched-risk point on that construction; like the 38.8%/16.6% pair it dates from the pre-dividend-adjusted generation, so treat it as the size of the effect rather than as a current headline figure.

05 — WHY TOTAL RETURN

Closing prices understate exactly the stocks a cautious book wants

A closing price omits dividends. A total-return (dividend-adjusted) series reinvests them. The choice is not neutral, and its bias is not random — it falls hardest on precisely the high-yield, low-volatility names a minimum-variance portfolio is made of.

MEASURED — SWITCHING TO TOTAL-RETURN PRICES

median expected return 7.80% → 10.39% · stocks with negative expected return 3 → 0 · correlation with the previous estimate 0.972

The *level* moved substantially; the *ranking* largely did not. The frontier shifts up by roughly 2.8 pp, and the gain lands hardest at the risk-averse end — which is where the high-dividend names sit.

Three independent checks that the adjusted file is correct

A supplied "adjusted" file has to be verified, not trusted. Three signs, each of which would fail if the adjustment were broken:

- **It is back-adjusted and anchored at the end.** The ratio adj/raw runs 0.68 → 0.81 → 1.00 for Unilever, a high payer, and 0.959 → 0.999 for NVDA, which pays almost nothing. That is the correct shape: the most recent price is unchanged and history is scaled down.
- **Volatility is unchanged** — 31.08% → 31.05%. Dividends add drift, not noise. A change here would mean the adjustment had corrupted the return series.
- **The uplift is ordered by dividend yield across sectors:** Energy +3.89 pp, Utilities +3.66, Financial Services +3.36, Real Estate +3.31, Consumer Defensive +2.76, against Technology +1.18 and Healthcare +1.23. Exactly the ranking dividend yields predict.

06 — EXPECTED RETURNS

Shrinking μ , because a raw historical mean is a trap

Expected return per stock is the annualised mean daily log return over the ten-year window, then **James–Stein shrunk** toward the cross-sectional average. The shrinkage is the part that needs defending.

A ten-year sample mean is a very noisy estimate of an expected return. That noise would be tolerable in a descriptive study, but a mean-variance optimiser is not a passive reader of its inputs: it actively searches for the highest-return assets, so it **preferentially selects whichever stocks the estimation error flattered**. The optimiser is, in effect, an error-maximiser.

THE EXACT ESTIMATOR

```
# target: ONE scalar, the grand mean of the raw estimates over the # 989 complete-history
stocks. Not segmented by region or sector. tau = mean(mu_raw) = 10.44% tau^2 = var(mu_raw)
cross-sectional variance se^2_i = sd^2_i / n_i each stock's own squared standard error w_i =
tau^2 / (tau^2 + se^2_i) weight on the stock's own estimate mu_i = w_i * mu_raw_i + (1 - w_i) * tau
the shrunk estimate
```

This is the empirical-Bayes form of the James–Stein estimator: the shrinkage intensity is *per stock* and is computed entirely from the data — **there is no hand-picked parameter anywhere in it**, no tuning constant and no choice of target beyond "the cross-sectional mean". A stock whose own estimate is noisy relative to the spread of estimates gets pulled hard toward the average; a well-measured one barely moves. Measured on the delivered inputs, the weight a stock keeps on its own estimate ranges from **0.030 to 0.702**, median 0.387 — so between 30% and 97% of each estimate comes from the common target.

MEASURED ON THE DELIVERED INPUTS — WHAT SHRINKAGE REMOVES

per-stock estimates: raw range -25.8% ... +119.1% → shrunk range +2.7% ... +19.1%
the frontier's maximum return: **62.63% raw → 17.63% shrunk**

A single stock with a 119% ten-year mean is not an opportunity, and a 62.6% frontier corner would not survive one question. Note what shrinkage does *not* do: it barely moves the ranking (correlation with the raw estimates stays high), so it is not overriding the data — it is removing the part of the spread that is attributable to estimation error rather than to differences between companies.

A STALE FIGURE, CORRECTED

The interim presentation and an earlier draft of this document quoted this effect as **38.8% → 16.6%**. Both of those numbers are real but they come from an *earlier data generation* — local-currency, non-dividend-adjusted prices with the pre-v4 expected returns (`pipeline/07_final_portfolio.py`). On the delivered inputs the pair is **62.63% → 17.63%**. The conclusion is unchanged and in fact stronger; only the figures were stale.

MEASURED — WHY A SHORTER WINDOW IS NOT THE FIX FOR ANYTHING

maximum frontier return: 38.8% on ten years → 85.6% on three

Shortening the estimation window was considered as a way to include recently-listed stocks. It inflates μ catastrophically: individual sample means reach 88% a year purely from the 2023–25 regime. The problem was solved a different way — see section 07.

07 — THE RISK MODEL

A 20-factor covariance, and what the 20 factors actually are

Why not a plain sample covariance matrix

The obvious estimator — the sample covariance of the return matrix — has a structural problem here. Every pairwise covariance needs a period in which *both* stocks traded, so a complete-case rule silently deletes any stock without a full history:

MEASURED — THE COST OF THE COMPLETE-CASE RULE

96 of 1,093 stocks excluded · the recovered names take 11.2% of capital at the min-risk end

Those 96 stocks could never be bought, no matter how attractive. And they were not marginal: once a factor model let them back in, they took over a tenth of the minimum-risk portfolio. Excluding a stock because of when it listed is not a modelling choice, it is a bug with a plausible-sounding name.

What a factor model does instead

Rather than estimating every pairwise covariance directly, estimate how each stock responds to a small set of common factors:

```
# factors extracted by PCA on the stocks that DO have a complete history # each stock's
betas fitted on whatever days that stock actually has  $\Sigma = B \cdot F \cdot B^T + \text{diag}(D)$  B betas,
one row per stock, 20 columns F covariance of the 20 factors D residual variance per
stock, on the diagonal
```

NO LEDOIT-WOLF ANYWHERE IN THE DELIVERED MODEL

covariance_matrix_v4.csv is **pure factor structure**: $\Sigma = B F B^T + \text{diag}(D)$ and nothing else. There is no Ledoit-Wolf step over it, under it, or blended into it — the shrinkage in the delivered model is on **μ only** (section 06). The factor structure is itself the regulariser for Σ : it replaces roughly 590,000 free pairwise covariances with $1,077 \times 20$ betas plus 1,077 residual variances.

Ledoit-Wolf appears only in the *backtests and the crisis stress test*, and only for the protocol reason: those re-estimate at 31 rebalance dates, and re-running the 20-factor pipeline that many times would change the estimator and the protocol in the same step, making the comparison uninterpretable. Anything quoted from those tests is a statement about a Ledoit-Wolf covariance; anything quoted from the static model is a statement about the 20-factor one. The two are never mixed inside one number.

Two properties follow, and both are the reason for the choice. **No pair of stocks ever needs a shared history**, so a 2022 listing costs the other 1,092 nothing. And **Σ is positive semi-definite by construction** — $B F B^T$ is a quadratic form and the diagonal is non-negative — which a quadratic optimiser requires to have a well-posed convex problem at all.

Why exactly 20 factors

The failure mode of a factor model is specific: it declares every off-diagonal *residual* covariance to be zero. If residuals actually co-move — same sector, same country — the model understates portfolio risk, and a minimum-variance optimiser will go looking for that blind spot. So the number of factors was chosen by measuring exactly that, sweeping k and comparing each portfolio's predicted risk against the risk those same weights actually realised:

FACTORS	VARIANCE EXPLAINED	MEDIAN R^2	ERROR AT MIN-RISK	VERDICT
1	27.8%	0.273	-38.34%	understates risk badly
2	35.4%	0.355	-37.55%	understates risk badly
5	42.8%	0.436	+0.31%	clears zero by a third of a point
10	47.5%	0.469	-1.56%	falls back below zero
20	52.9%	0.518	+1.86%	first non-marginal margin — chosen
30	56.4%	0.547	+3.06%	safe but no gain
50	61.4%	0.586	+5.90%	overstates; condition number worsens

Error = predicted risk ÷ realised risk - 1. Negative means the model understates risk. An in-sample consistency check: it asks whether the optimiser can game a covariance that assumes diagonal residuals, not whether the covariance predicts the future.

One or two factors understate portfolio risk by **38%** at the min-risk end. That is a cliff, not a gradient, and it is the whole argument against a single-index model. Above five factors the understatement disappears — but the boundary is unstable: five clears zero by 0.31% and ten falls back to -1.56%. That non-monotonicity is the metric working as designed, because each k produces a different portfolio, so what is measured is whether the optimiser can find a blind spot in that particular matrix, and near zero the answer is noisy.

HOW THE CHOICE IS STATED

20 is not the minimum — it is the smallest count with a margin wide enough not to depend on which portfolio the optimiser happens to pick. More than 20 only inflates the overstatement and the condition number (2,957 → 4,389 from $k=5$ to $k=50$) while median R^2 climbs from 0.436 to 0.586, which past some point is fitting noise into the common structure instead of removing it from the residual.

What the factors are, since nobody named them

They are principal components — eigenvectors of the return covariance — not factors chosen in advance. Nothing about them is specified by us except how many to keep. They are nevertheless interpretable after the fact:

FACTOR	VARIANCE	WHAT IT TURNS OUT TO BE	EVIDENCE
1	27.8%	the market	correlation 0.996 with an equal-weight index of the same stocks
2	7.5%	Europe vs the United States	mean beta +1.43 European, -1.73 US
3	3.4%	Energy vs Technology	mean beta -2.76 Energy ... +1.75 Technology
4	2.4%	Technology vs Utilities	mean beta -1.47 Technology ... +2.94 Utilities
5-20	0.4-1.6% each	no single clean reading	the tail that carries the residual block structure

Factor 2 settles an argument. An **explicit** Europe/US factor was added to the risk model earlier as an attempted fix: it absorbed the region structure completely (residual-correlation spread +0.2237 → -0.0010) and left the risk understatement essentially unchanged, 13.6% → 13.1%. The reason is now visible — PCA finds the region split by itself, as the second component, without being told. **The failure was never which factor is named. It is how much variance is left inside a residual block the model declares diagonal.**

A mixed-integer quadratic programme, and three rejected alternatives

minimize $w^T \Sigma w$ portfolio variance subject to $w^T \mu \geq \beta$ target return – swept to trace the frontier $\sum w_i = 1$ full investment $0.01 \cdot y_i \leq w_i \leq 0.20 \cdot y_i$ 1–20% if held, exactly 0 if not $y_i \in \{0,1\}$ one binary per stock: held or not $\sum y_i \geq 30$ minimum holdings + region, sector, ESG and screening constraints (section 09)

Solved in FICO Xpress. 1,077 continuous variables, 1,077 binaries, roughly 1.16 million quadratic elements in the objective. **0.3–1.1 seconds per solve** at a 0.1% MIP gap, which is what makes a 15-point frontier and a 15-scenario grid practical.

Why the objective is quadratic, not a linear risk proxy

A common simplification replaces the quadratic form with a linear risk measure — a weighted sum of per-stock volatilities, or a mean absolute deviation. It is tempting because it turns an MIQP into a MILP. It is also wrong here, because a linear measure cannot see *correlation*: it treats two perfectly correlated stocks as diversification.

WHERE MODEL 1 IS – IT IS A REAL MODEL, NOT JUST A COMPARISON

The case study offers two routes to portfolio risk and invites a comparison, so the linear one was **implemented as its own model and solved**, not merely reasoned about. `archive/our_model_frozen/15_model1_linear.py` is a 127-line MILP minimising $\sum w_i \sigma_i$ over the *identical* constraint set, and it verifies that claim rather than asserting it: the same problem builder is first run with the quadratic objective and checked against the MIQP's own solution, so the two models provably differ only in the objective.

Two things to be straight about. It lives in the frozen archive and was run on the *earlier* data generation, so it is not maintained against the current v4 inputs. And $\sum w_i \sigma_i$ is the portfolio standard deviation you would get if every pair of stocks were perfectly correlated — an upper bound on true risk that is completely blind to diversification, since swapping a stock for another of the same σ never changes the objective however differently the two move.

MEASURED – THE COST OF LINEARISING RISK

the linear model's book carries +19.2% more true risk · its objective/actual gap is 1.64×

The linear model does not merely perform worse, it *misreports itself* by a factor of 1.64. Diversification is a covariance phenomenon, so a risk measure that ignores covariance is measuring something other than risk.

Why binary variables, and not semi-continuous ones

Xpress supports semi-continuous variables, which express "either zero or between 1% and 20%" directly and would remove 1,077 binaries. Tested, and rejected:

MEASURED – SEMI-CONTINUOUS VARIABLES ALONE

returned 26 real positions while reporting 30

The "at least 30 stocks" condition has to count something. With explicit binaries it counts $\sum y_i$. With semi-continuous variables alone there is no counter, and the constraint silently attaches to something that is not the number of holdings — so the model satisfies a condition the portfolio violates. A wrong answer that validates is worse than a slow one.

Why the frontier is traced by sweeping β , not by weighting two objectives

The textbook alternative is a weighted sum: minimise $\lambda \cdot w^T \Sigma w - (1-\lambda) \cdot w^T \mu$ and sweep λ . FICO's own example notebook does exactly this. It is wrong for *this* problem:

- **A weighted sum only reaches points on the convex hull** of the Pareto frontier. With binary variables the feasible set is non-convex, so the frontier is non-convex, and some Pareto-optimal portfolios are unreachable by *any* weight. The ε -constraint form used here — minimise risk subject to return $\geq \beta$ — reaches all of them. The notebook's example has five assets and no binaries, where the distinction never appears.
- **Xpress' native multi-objective machinery is unusable here**, for a concrete reason. It requires every objective to be linear, which forces the variance into a transfer variable and a constraint. Measured: that converts a convex MIQP (1,159,929 quadratic elements in the objective, zero quadratic constraints) into a quadratically *constrained* MIP (580,503 quadratic elements, one quadratic constraint). The second form did not finish a single solve in over two minutes, against roughly 0.8 seconds for the first.

Why the MIP gap is 0.1% and not 1%

MEASURED — GAP TIGHTENED FROM 1% TO 0.1%

the measured cost of a cheap constraint read 0.078 pp at a 1% gap, against 0.042 pp at 0.1% and at 0.01%

At a loose gap the *measurement noise* was roughly the same size as the effects being measured — a constraint's cost read about double its true value. Since much of this project is measuring small differences between scenarios, the gap has to be tighter than the effects. It costs nothing: 0.4 seconds either way.

09 — CONSTRAINTS

What the brief gave us, what we added, and what each addition costs

GIVEN BY THE BRIEF

- $1\% \leq \text{weight} \leq 20\%$ per stock
- at least 30 different stocks
- any single region $\leq 60\%$
- weighted ESG ≥ 70

ADDED BY US

- any single sector $\leq 30\%$
- per-stock ESG floor ≥ 30
- Tier 1 controversial-weapons exclusion
- Tier 2 defence exclusion, as a toggle
- any single country $\leq 25\%$ — built, ships off

Sector cap of 30% — a guard-rail, and it is honest about not binding

The brief caps regions but not sectors, which leaves the optimiser free to concentrate entirely within one industry. A cap was added as a guard-rail. What it actually costs was then measured rather than asserted, at four levels:

SECTOR CAP	NEUTRAL RETURN	NEUTRAL RISK	RET/RISK	TOP SECTOR
none	14.319%	10.636%	1.3463	27.10%
20%	14.139%	10.548%	1.3404	20.00%
25%	14.290%	10.618%	1.3459	25.00%
30% — shipped	14.300%	10.622%	1.3463	27.18%

Read the top-sector column: uncapped, the recommended book puts 27.10% in its largest sector — **under the 30% limit, so at the recommended profile the cap does not bind at all** and costs 0.020 pp. Where it does bind is the *minimum-risk* end, where the uncapped book puts 33.95% into Consumer Defensive. Tightening to 20% starts to cost real money: −0.160 pp of return at Neutral, and it binds at every profile.

The cap also *lowers* risk where it binds — switching it off raises the risk-prone book's volatility from 22.63% to 23.14% — and it changes which frontier point qualifies as Risk Prone, for reasons explained in section 10.

Per-stock ESG floor of 30 — because a weighted average can hide anything

The brief's ESG condition is on the portfolio *average*. An average of 70 is satisfiable by holding a stock with an ESG score of 3 alongside enough high scorers. For a fund whose entire premise is sustainability, that is a reputational exposure rather than a mathematical one. A floor was added, and it is nearly free:

MEASURED — ESG COST, DECOMPOSED

the weighted-average ≥ 70 requirement costs −0.151 pp at Neutral · the per-stock floor costs +0.008 pp, i.e. nothing · floor at 20 / 30 / 40 gives 14.299% / 14.300% / 14.305%

The floor's exact level is not a load-bearing choice — the answer barely moves across 20, 30 and 40 — which is worth knowing, because an arbitrary-looking parameter that turns out not to matter is much easier to defend than one that does.

Weapons screens — because ESG scores do not screen weapons

MEASURED — WHAT THE ESG CONSTRAINT ALONE PERMITS

Rheinmetall carries an ESG score of 87.3 and takes 10.2% of the book without an explicit screen

An ESG rating measures governance and disclosure quality, not what a company makes. A well-run munitions manufacturer scores well. Presenting an "ESG-constrained sustainable portfolio" with 10% in controversial weapons would be indefensible, and no ESG threshold fixes it, because the score is not measuring that.

Two tiers, deliberately separated. **Tier 1** is the narrow, near-consensus case: companies whose overwhelming core business is weapons and ammunition, following Norwegian sovereign-fund precedent. Always excluded. **Tier 2** is diversified defence and aerospace conglomerates, where defence is a major but not sole segment — RTX also makes civilian jet engines, Kongsberg has maritime and digital divisions. That category is genuinely contested post-2022, so it is a *toggle* rather than a baked-in judgement, which lets the report show the exact cost of the policy choice instead of hiding it. The cost is **−0.004 pp** at Neutral.

A related defect is worth recording as a warning: four of the ten Tier 2 tickers did not exist in this dataset, so the screen was silently excluding six of ten. The dangerous one was **BA.L** for BAE Systems — **BA.N** exists in this dataset and is **Boeing**, so a plausible-looking correction would have excluded the wrong company.

Country cap of 25% — built, measured, and shipped switched off

The region cap constrains Europe as a whole. It cannot see a single country, and a sector cap cannot see a country that spans two sectors. Switzerland exploited both gaps at once:

MEASURED — CONCENTRATION NOTHING WAS RESTRAINING

Switzerland is 46.98% of the risk-averse book and the largest country at 14 of 15 frontier points · Switzerland + United States = 92.2% of that book

Cantonal banks and real estate — two different sectors, one small market. 92% of \$100 million in two countries, while passing every constraint the model had.

A 25% cap costs +0.112 pp of risk at Neutral at matched return, or -0.067 pp of return at matched risk, and removes 11.7 pp of Swiss concentration. It is expensive only at the extreme left of the frontier (+0.146 pp of risk = -1.39 pp of return at matched risk), because the frontier is roughly ten times steeper there. No cap level between 20% and 40% is ever infeasible.

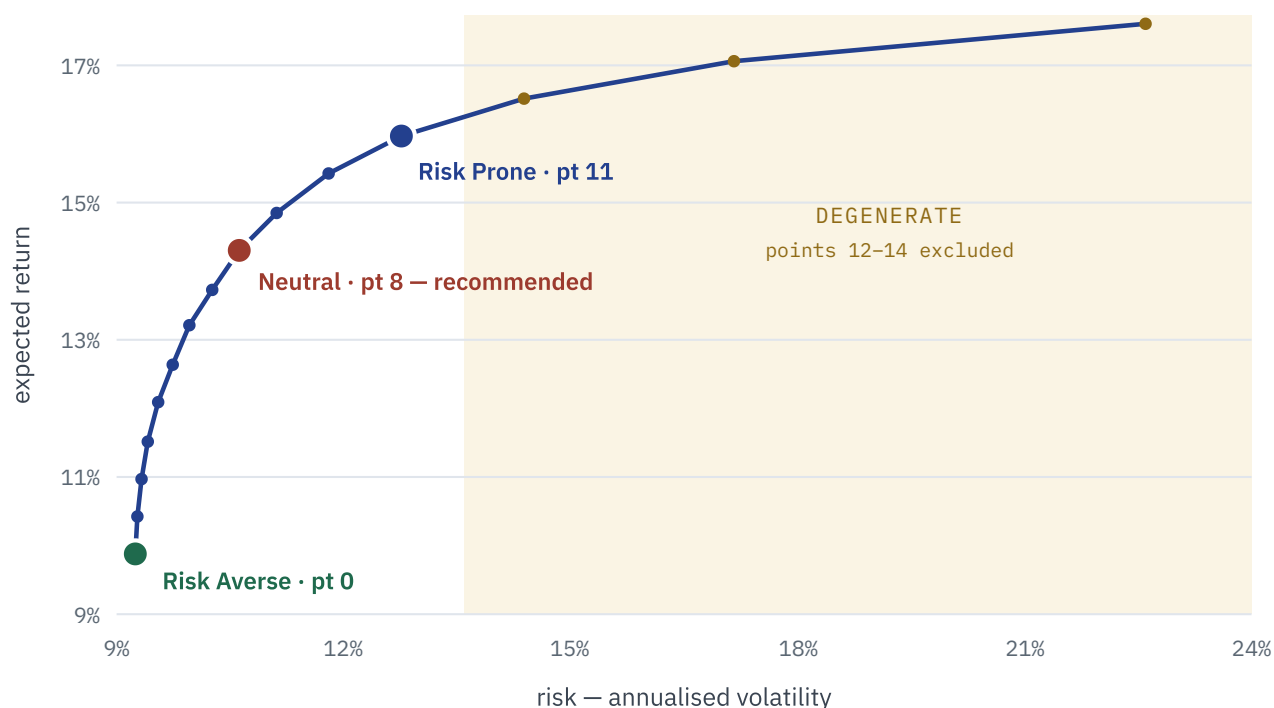
WHY THE UNITED STATES IS EXEMPT — FORCED, NOT CHOSEN

The brief's own region condition puts at least 40% in each region, so a country cap below 40% applied to the US is infeasible on its own. This is not a judgement call, and the solver proves it: removing the exemption at a 25% cap produces an irreducible infeasible set of exactly three constraints — the budget, the Europe region cap and the US country cap — with a total infeasibility of **0.15, which is exactly $1 - 0.60 - 0.25$** . 15% of the budget has nowhere to go.

10 — THE THREE PROFILES

Choosing three portfolios off a frontier of fifteen

The optimiser does not produce one portfolio; it produces a frontier. β is swept in 15 equal steps between the two corners — the global minimum-variance portfolio and the maximum return achievable — and each step is a separate MIQP solve. Three points are then reported, matching the brief's request for scenarios.



Fifteen solved portfolios. The frontier is steep on the left and flat on the right: past roughly 14% risk, extra risk buys very little extra return. That shape is the argument for not choosing the right-hand corner.

The selection rule, and the inconsistency it resolved

Two definitions are straightforward. The third is not, and the project spent real time getting it wrong:

- **Risk Averse** = minimum variance. The left-hand corner.
- **Neutral** = maximum return/risk ratio. The best risk-adjusted point on the frontier, with no arbitrary target and no risk-aversion parameter to pick.
- **Risk Prone** = highest return among **non-degenerate** points, where degenerate means the top three holdings exceed 40% of the budget *or* more than 15 positions sit on the 1% floor.

"Maximum return" without the degeneracy filter lands on a corner of the feasible set rather than a portfolio. At point 14, three positions are pinned at the 20% cap, 23 of 30 positions sit on the 1% floor, and 60% of the budget is in three stocks. Its return/risk ratio of 0.779 is *worse* than the risk-averse book's 1.067 — it is a dominated portfolio that no mandate holder would sign.

POINT	RETURN	RISK	RET/RISK	TOP-3 WEIGHT	ON THE 1% FLOOR	STATUS
11	15.97%	12.77%	1.250	33.8%	10	Risk Prone
12	16.52%	14.38%	1.146	34.0%	16	degenerate — floor count
13	17.08%	17.15%	1.000	40.5%	21	degenerate — both tests
14	17.63%	22.62%	0.779	60.0%	23	degenerate — the corner

WHERE 40% AND 15 COME FROM

Both thresholds were originally a **judgement call, not a measurement** — they appear as literals in the frozen script with no sweep behind them. Since the choice now decides which portfolio is called Risk Prone, they have been swept over a 9 × 9 grid, and the result is more defensible than the original reasoning:

- **The 40% top-3 test is inert on this frontier.** Varying it from 34% to no limit at all changes the selected point only when the floor test is simultaneously loosened past 22. It is a belt-and-braces condition that never binds here.
- **The floor count is what decides, and point 11 is selected for any threshold from 10 to 15.** Point 11 has 10 positions at the floor and point 12 has 16, so any value in that six-wide band gives the same answer. Below 10 the pick moves to point 10; at 18 and above it moves to 12.
- **15 has a clean reading at the holding counts that matter.** Every point from 8 rightward holds exactly 30 stocks, so "more than 15 at the floor" is exactly "*more than half the book sitting at the minimum weight*". That is a statement about the optimiser padding to satisfy the 30-stock count rather than choosing 30 stocks — which is a reason, not a tuned number.

So the honest answer if challenged: the thresholds were set by judgement, the selection is stable across the whole plausible range of the one that matters, and 15 is the loosest value in that stable band with an interpretation that does not depend on the number itself.

A REAL INCONSISTENCY, FOUND AND FIXED

For a period the repository labelled *two different portfolios* "Risk Prone": the holdings files used point 11, while the profile definitions, the scenario matrix and the presentation deck used point 14. The rule above already existed in a frozen script and was simply not applied everywhere. It now lives in `profile_rule.py` and nowhere else, and is *derived* in every output rather than hard-coded — which matters, because the degenerate set moves with the constraints: with no sector cap, Risk Prone shifts to point 12.

One artefact was never realigned: the interim presentation still shows point 14 (17.63% / 22.62% / ratio 0.779). The presentation has since been retired as a deliverable, so it was left alone rather than rebuilt. If any of those slides is shown, it will contradict every current output — **the number to quote is point 11, 15.97% at 12.77%, ratio 1.250.**

The delivered portfolios

PROFILE	RETURN	RISK	RET/RISK	HOLDINGS	WEIGHTED ESG
Risk Averse	9.87%	9.26%	1.067	42	70.00
Neutral — recommended	14.30%	10.62%	1.346	30	70.00
Risk Prone	15.97%	12.77%	1.250	30	70.00

Weighted ESG sits at exactly 70.00 at every one of the 15 frontier points, meaning the ESG constraint is always active — and, per section 09, costs almost nothing.

11 — TESTS

Nine tests, why each was run, and what it showed

The governing principle: every non-obvious choice is measured rather than asserted, and every rebuild is gated against something it must reproduce. A number that cannot be reproduced is not evidence.

1. Scenario matrix — what each constraint costs

Why: a constraint whose price is unknown cannot be defended or relaxed. Eight constraint configurations × three profiles, all solved. Produces the ESG cost decomposition, the Tier 2 cost, the sector-cap effect and the ESG-floor insensitivity quoted in section 09.

2. Factor count — how many factors the risk model needs

Why: the single most arbitrary-looking parameter in the model. Section 07. Re-run on the current data after the inputs changed, because the original measurement used non-dividend-adjusted prices and a superseded μ — the number 20 had been inherited, not re-derived. *Gated:* the rebuilt covariance at $k=20$ reproduces the shipped matrix to 1.8×10^{-15} .

3. Country cap — the concentration nothing restrained

Why: Switzerland at 47%. Section 09. Four cap levels swept across the full frontier, cost reported in both framings (risk at matched return, and return at matched risk) because they differ by an order of magnitude.

4. Walk-forward backtest — does the optimiser work out of sample?

Why: everything above is in-sample. A rolling three-year estimation window, quarterly rebalancing, 31 rebalances over 7.7 years, buy-and-hold in between so weights drift as a real book does. At each rebalance date only data strictly *before* it is used. A deliberately simpler estimator is used here — trailing mean log return and a Ledoit-Wolf shrunk sample covariance — because re-running the full 20-factor pipeline 31 times would change the estimator and the protocol at once, making the comparison meaningless.

5. Profile backtest — testing the book we actually recommend

Why: test 4 optimises for minimum variance at every rebalance, so it validates the *risk-averse* end, not the recommendation. Four books were run on the identical protocol. Expressing a return-seeking profile required care: an absolute return target is not comparable across regimes (the same β is easy in 2021 and infeasible in 2018), so the profile is stated as a **mandate** — "maximise return, then minimise risk while giving up at most X of the *achievable* maximum" — which is defined relative to the regime and costs two solves rather than a whole frontier. *Gated:* the minimum-variance book reproduces test 4 to 0.006 of Sharpe.

6. The option-B check — testing a methodology choice

Why: a simpler protocol was proposed — split history into three chunks and rebalance twice. It was declined on the argument that two out-of-sample observations cannot separate skill from luck. That argument was then *tested* rather than left as an assertion: the same protocol was run with two different, equally defensible split dates.

MEASURED — THE TWO-REBALANCE PROTOCOL CONTRADICTS ITSELF

the recommended profile "beats 1/N" ($\Delta\text{Sharpe} +0.087$) on one split and "loses to 1/N" (-0.244) on another · spread 0.331 · the 31-rebalance answer is $+0.010$

Decided by nothing but where the boundary falls. It would have produced a confident-looking result in either direction from the same data and the same code. With so few rebalance points the test window moves with the split too, so sample size and window choice are not separable — a second reason the protocol is fragile here.

7. Crisis stress test — how the book behaves when markets break

Why: average statistics say nothing about the tail. Two panels, deliberately kept apart. **Panel A** holds the delivered portfolios through seven historical crisis windows; it is *in-sample*, because μ and Σ were estimated over a period containing all of them, so it describes the delivered book but proves nothing. **Panel B** estimates on the three years strictly before each window, re-solves, and holds through — that is the actual test. The windows were also checked against the five worst 60-day stretches found empirically, to confirm they were not cherry-picked.

8. The 1/N benchmark, over all eleven years

Why: "our portfolio returned 14.3%" means nothing without a reference for what no skill would have earned. Five variants, because 1/N is not one thing. *Gated:* agrees with an independently-built 1/N to 0.001 of Sharpe.

MEASURED – TWO THINGS THAT CHANGE HOW THE BENCHMARK MUST BE PRESENTED

rebalancing frequency moves it by 0.9 pp of CAGR (daily 15.67%, buy-and-hold 15.35%, monthly 14.80%) · no 1/N variant is admissible under the brief

Equal weight over 1,077 names puts 0.093% in each, against a 1% floor, and no equal-weight portfolio over more than 100 names can satisfy that floor at all. The unscreened variants also miss weighted ESG ≥ 70 ; the ESG-screened one clears ESG and then breaks the 60% region cap, because the high-ESG half of the universe is 63% European. 1/N is a reference, not a portfolio we could have proposed — and any comparison has to say which 1/N it means.

9. Estimation-window robustness — 10, 5 and 3 years

Why: the window length is the input the model is least robust to, and it is chosen by us. μ and Σ are *re-estimated* per window, not merely re-solved, because the window is the input being varied. *Gated:* at ten years the rebuild reproduces the shipped estimates to 1.7×10^{-16} (μ) and 1.8×10^{-15} (Σ), so the shorter windows measure the window rather than the rebuild.

MEASURED – SHORTER WINDOWS LOOK BETTER AND ARE WORSE

Neutral: 14.30% / 10.63% at ratio 1.35 on ten years → 17.47% / 9.32% at 1.87 on five → 25.73% / 7.93% at **3.24** on three · μ 's range widens from 2.7–19.1% to –9.8–46.1%

Nothing improved. The three-year estimate claims the book earns 25.7% at 7.9% risk; that is the noise in a three-year mean, and the optimiser is buying whatever the noise flattered. If a dashboard shows these side by side, the short windows must be labelled as estimation error, not as an alternative recommendation.

Also run: the second LSEG data export

A refreshed ESG column plus GICS sector and industry classifications arrived late. It was parsed and measured rather than adopted. Adopting both changes moves the recommended book by **0.069 pp**, which is itself a useful robustness result: an independent, newer data pull barely moves the answer.

The stated reason for not adopting it was that its ESG column is **not one cross-section** — the per-cell formulas show fiscal period FY2024 for 556 tickers and FY2025 for 529, plus a handful of FY2022, FY2023 and FY2026 — so an ESG ≥ 70 constraint on it compares one company's 2024 score with another's 2025.

THAT REASONING NEEDS A CAVEAT, AND HERE IT IS

The obvious challenge is whether *our own* column has the same problem, which would make the objection self-defeating. Honest position: **we cannot verify our own column's vintage**. It arrived as a plain CSV in the course dataset with no per-cell metadata, so unlike the second export — whose mixing is visible because the file carries its own formulas — ours is simply unknown.

An indirect test is possible and was run. If our column were mixed in the same pattern as theirs, its distance to their values should differ systematically between their FY2024-tagged and FY2025-tagged tickers. It does not: mean absolute gap 1.73 points on the FY2024 group against 1.98 on the FY2025 group, a difference of 0.24 points at $t = -1.52$, not significant. So our column is roughly equidistant from both, consistent with it being its own single earlier vintage rather than a mix aligned with theirs. That is evidence, not proof, and the test has limited power.

Which means the sharper framing is this: the real trade is **a column of unverifiable but probably uniform provenance containing 49 invented values, against a column of real values whose mixing is documented**. Put that way, "verifiably mixed" is not obviously worse than "unverifiable", and the recommendation in section 14 leans toward adopting theirs and stating the fiscal-period spread as a known limitation — because a documented limitation is easier to defend than an assumption you cannot check.

12 — WHAT FAILED

The negative results, stated as findings

Four things did not work. They are reported here rather than buried, because a study that only produces confirmations has not tested anything — and because each of them changes what can honestly be claimed.

1 — NO PORTFOLIO BEATS THE NAIVE BENCHMARK ON RISK-ADJUSTED RETURN

Sharpe against 1/N over 31 rebalances: minimum variance -0.038 ($t = -0.87$), mandate-20 $+0.010$ ($t = +0.60$), mandate-10 $+0.047$ ($t = +1.12$)

Every t-statistic sits inside ± 1.96 . The whole Sharpe spread from 0.829 to 0.914 is noise, so no risk-adjusted claim is available *in either direction* — including the negative one. Over 7.7 years the honest answer is "not distinguishable from the benchmark". The apparent edge of the return-seeking books also does not survive costs: they turn over 170% a year against minimum variance's 83%, and at 10 basis points the mandate's Sharpe falls to 0.857, below the benchmark's 0.866.

2 — THE MODEL UNDERSTATES ITS OWN RISK OUT OF SAMPLE, WORST WHEN IT MATTERS MOST

predicted 8.78% vs 13.21% realised across the backtest — a 50.4% understatement · in crises, realised ÷ predicted was 1.83× in 2018 Q4, 1.44× in 2022 and **6.71×** in the COVID crash

The 50% average understatement is the ordinary case; the tail is far worse. The static model quotes 10.62% risk for the recommended book, and in a crash the same construction realised nearly seven times its own forecast. Any risk number this model produces has to be presented as a lower bound.

3 – THE PORTFOLIO IS UNSTABLE ACROSS ESTIMATION WINDOWS, AND IT IS NOT NOISE

against the ten-year book, the five-year Neutral shares 15 of 38 names (Jaccard 0.283, 64.5% of capital placed differently); the three-year shares 11 of 38 with weight correlation -0.019

Two follow-ups establish that this is regime change rather than sampling error: bootstrap resampling *within* a window converges to a stable core (top selection frequency 1.00), while cores from different windows disagree — and resampling as a remedy moves the overlap only from 0.29 to 0.31. The instability is real information about the market, not a defect to be smoothed away.

4 – THE RECOMMENDED PROFILE IS NOT WHERE THE ROBUSTNESS LIVES

out of sample the recommended book beats 1/N in one crisis of three (COVID +6.1 pp, 2018 Q4 -0.8 pp, 2022 -6.8 pp) while the minimum-variance book beats it in all three (+7.6 / +4.4 / +2.1 pp)

Three independent analyses now point the same way. The crisis stress test, the walk-forward backtest and the window-stability study each single out the *minimum-variance* end as the robust one. Neutral remains the best risk-adjusted point on the *estimated* frontier, which is a statement about the estimate rather than about the future.

WHAT THIS IMPLIES FOR THE RECOMMENDATION

The defensible claim is about **risk, not return**: recommend on measured volatility and drawdown reduction — the backtested volatility edge is -22% overall, lower in five of six sub-periods, winning both the 2020 crash and 2022 — and say plainly that no risk-adjusted edge is statistically demonstrable over 7.7 years. That is a stronger position at a viva than a Sharpe ratio that does not survive a t-test.

Also honestly negative, in the smaller print

- **Turnover.** 20.8% per rebalance, roughly 83% a year, costing 0.18 pp of CAGR at 10 basis points.
- **Country and sector caps do not substitute for one another.** With only a country cap, one sector reaches 39.8%; with only a sector cap, Switzerland reaches 46.8%. Both are needed, or neither claim can be made.
- **The naïve ESG screen is worse than the constraint, but by less than it looks.** On the strictly comparable measure — expected return under the same μ — screening out everything below ESG 70 costs 0.263 pp against the weighted-average constraint's 0.151 pp, a factor of 1.7. The realised eleven-year gap is 1.47 pp, 5.6× larger than the ex-ante gap, and that difference is not in μ : it comes from the exposures the screen imposes (63% Europe), not from picking lower-return stocks.

13 – REPRODUCIBILITY

Gates, and the defects they caught

The brief asks that external data be documented and reproducible. In practice that principle had to be extended to the analysis itself, because several numbers turned out not to be reproducible at all. The working rule: **every rebuild is gated against something it must reproduce, and the gate is reported next to the result.**

GATE	RESULT
Rewriting the group caps in a different idiom	byte-identical LP file; all 15 frontier points re-solved with max $ \Delta \text{ weight} = 0.000$
Rebuilding the covariance at $k=20$	reproduces the shipped matrix to 1.8×10^{-15}
Re-estimating on the ten-year window	reproduces the shipped μ to 1.7×10^{-16} and Σ to 1.8×10^{-15}
New backtest harness vs the original	Sharpe 0.829 vs 0.835, volatility 13.197% vs 13.210%
New 1/N vs an independently built one	Sharpe 0.867 vs 0.866
Regenerated presentation inputs	every field matches the original; the rebuilt PDF is text-identical

Three defects the gates exposed

- **The presentation was unbuildable by anyone else.** The deck-building script read a JSON file from `/tmp` that *nothing in the repository wrote* — it had been produced by an ad-hoc snippet in an earlier session. Every scenario number on the slides came from it, and macOS clears `/tmp` on reboot, so the documented rebuild command crashed on a clean checkout. It is now regenerated from committed artefacts only, verified field-by-field against the original, and a clean clone with no licence file and no `/tmp` contents builds a text-identical deck.
- **Two backtest inputs were read but never written** by any script, for the same reason. Regenerable replacements now exist.
- **A subtle tolerance bug.** Counting positions "at the 1% floor" with a tolerance of 10^{-6} reports zero where there is plainly one, because at a 0.1% MIP gap the solver leaves the smallest holding at 0.010005 — five parts per million above the bound. The correct tolerance is 10^{-5} , and this feeds directly into the degeneracy test in section 10.

A REPRODUCIBILITY LIMIT WORTH STATING PLAINLY

The stored frontier is not bit-reproducible at its flat left-hand end: points 0 through 6 differ by up to 7.7×10^{-5} in return and by one holding. The pre-existing code reproduces the *same* deviation exactly, so this is not a code change — it is the 0.1% MIP gap admitting genuinely alternative optima where the frontier is nearly flat and many portfolios are almost equally good. Points 8 through 14, including the recommendation, match to the last digit.

14 — STILL OPEN

What is not done, and what it would take

1. **Whether to switch the country cap on.** Built, measured, ships off. 25% is the recommendation: 40% barely binds, and 20% costs 44% more for 5 pp less concentration. Turning it on means re-running the scenario matrix and the deck.
2. **Whether to adopt the second LSEG data export.** A provenance decision, not a results decision — see section 11. The recommendation is to take the GICS sectors unconditionally and the ESG column *whole* rather than blending two pulls, stating the mixed fiscal period as a known limitation.
3. **Aligning the presentation with the canonical Risk Prone.** The deck still shows the degenerate point 14. A one-line change plus a rebuild, deliberately left for a decision because it changes a slide.
4. **Re-extracting the six missing price series.** Currently dropped rather than back-filled.
5. **Factor-level return attribution and transaction costs inside the optimiser.** Turnover is measured after the fact but never priced during optimisation, so the model has no incentive to prefer a cheaper rebalance.

6. **A shrunk- μ variant of the walk-forward mandate.** Out of sample the mandate anchors on the max-return corner of an *unshrunk* three-year sample mean — it predicted 45.85% return at the first rebalance. Shrinking μ per rebalance would separate "the profile fails out of sample" from "an unshrunk three-year mean fails out of sample", at no extra solve cost.

KNOWN LIMITATIONS THAT CANNOT BE FIXED WITH THIS DATASET

Three, inherited by every backtested and stress-tested number here. **ESG scores and sectors are a single present-day snapshot**, so applying $\text{ESG} \geq 70$ at a 2018 rebalance uses 2025 information — look-ahead that cannot be removed without a historical ESG panel. **The universe is today's index constituents**, so it is survivorship-biased by construction: companies that failed or were delisted are absent, and realised returns are therefore optimistic for every strategy, benchmark included. **Transaction costs are not in the optimisation.** All three are reported rather than hidden.

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Every figure in this document is reproducible from the scripts named in the project handoff. Percentage points are written pp; basis points, bp. Model: MIQP in FICO Xpress 9.9.1, optimizer 47.01.02, 0.1% MIP gap.